



Ishan S. Khare

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EDUCATION

Stanford University

Stanford, CA

M.S. Computer Science

Jun 2025 – Dec 2026

- **GPA:** 3.96/4.0; NSF Graduate Research Fellow.
- **Teaching Assistant:** NLP, Computer Vision, Decision Making Under Uncertainty

B.S. Computer Science, with Distinction

Sep 2021 – Jun 2025

- **GPA:** 4.04/4.0; **Honors:** *Tau Beta Pi, Phi Beta Kappa*.
- **Selected Coursework:** Information Theory, Statistical Inference, Combinatorics, Modern Algorithms, Parallel Computing, Deep Reinforcement Learning, ML from Human Preferences.

EXPERIENCE

Stanford Artificial Intelligence Lab

Dec 2023 – present

Graduate AI/ML Researcher

Stanford, CA

- Conduct research with Profs. Chris Ré (Hazy Research) and Azalia Mirhoseini (Scaling Intelligence) on efficient and scalable language models.
- Co-developed Next-Concept Modeling (NCM) for coarse-to-fine text generation that runs **21× faster than diffusion baselines** on LM1B at better Gen. PPL, with **5.5× fewer training tokens**.
- Interested in identifying the fundamental bottlenecks and principles governing modern ML systems, then using those insights to guide more efficient and capable designs.

IMC Financial Markets

Jun 2024 – Aug 2024

Quantitative Trader Intern

Chicago, IL

- Led a project with traders and PhD researchers to develop deep learning methods for short-horizon index options prediction and portfolio allocation.

Stanford CS Theory Group

Jun 2023 – Dec 2023

Algorithms Research Assistant

Stanford, CA

- Selected for Stanford's CURIS program and advised by Profs. Aviad Rubinfeld & Moses Charikar.
- Contributed to LP/Lagrangian relaxation techniques for improved approximation algorithms for continuous Euclidean k -means.

PUBLICATIONS

- **Next-Concept Modeling: Fast, Parallel Coarse-to-Fine Text Generation**
H. Kumbong*, I.S. Khare*, ..., S. Arora, A. Mirhoseini, C. Ré. *Under Review*. *Equal first-author.
- **An Information Theoretic Perspective on Agentic System Design**
S. He, A. Narayan, I.S. Khare, C. Ré, S. Linderman, D. Biderman. *ICLR 2026*.
- **Smoothie: Label Free Language Model Routing**
N. Guha, M.F. Chen, T. Chow, I.S. Khare, C. Ré. *NeurIPS 2024*.
- **Wonderbread: A Benchmark for Evaluating Multimodal Foundation Models on Business Process Management Tasks**
M. Wornow, A. Narayan, B. Viggiano, I.S. Khare, ..., C. Ré. *NeurIPS 2024*.
- **Block Parallelism for Efficient Distributed Long-Context Diffusion Language Model Training**
T. Suresh*, P. Chaturvedi*, H. Kang*, P. Shroff, I.S. Khare, H. Kumbong, A. Mirhoseini. *NeurIPS DiffuLM & LCFM Workshops, 2026*.
- **Electronic, Optical, and Thermoelectric Properties of Sodium Pnictogen Chalcogenides**
I.S. Khare, N.J. Szymanski, D. Gall, R.E. Irving. *Computational Materials Science, 2020*.

SELECTED HONORS

NSF and NDSEG Fellowships; Research Science Institute Scholar (top 52 in USA); AIME Qualifier; Regeneron Science Talent Search Scholar; U.S. Chem Olympiad National Finalist; Coca-Cola Scholar; National Merit Scholar; Eagle Scout